

Geometry and Measures

GCSE Maths · Practice Questions

20 questions · 62 marks total

jtmaths.org · free maths resources

Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Two angles lie on a straight line. One of them is 130° .
Work out the size of the other angle.

.....

2 [2 marks]

- (a) Work out the area of a triangle with base 10 cm and perpendicular height 6 cm.
(b) Work out the area of a parallelogram with base 7 cm and perpendicular height 4 cm.
-

3 [2 marks]

Two angles of a triangle are 55° and 68° .
Work out the size of the third angle.

.....

4 [2 marks]

A circle has radius 5 cm.
Work out the circumference of the circle, correct to 1 decimal place.

.....

5 [3 marks]

A circle has diameter 12 cm.
Work out the area of the circle, correct to 1 decimal place.

.....

6

[3 marks]

Work out the size of one interior angle of a regular octagon.

7

[3 marks]

A right-angled triangle has the two shorter sides 5 cm and 12 cm.

(a) Work out the length of the hypotenuse.

(b) Work out the perimeter of the triangle.

8

[3 marks]

In the diagram, AB is parallel to CD. The line PQ crosses AB at X and CD at Y.

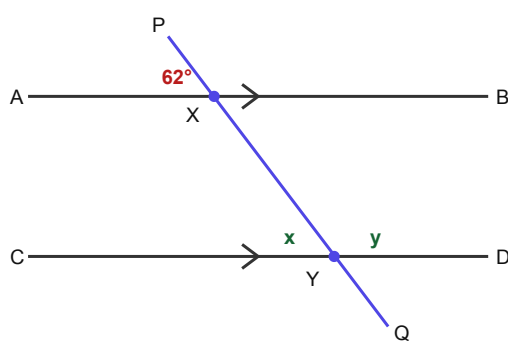


Diagram not accurately drawn

(a) Work out the size of the angle marked x . Give a reason for your answer.

(b) Work out the size of the angle marked y . Give a reason for your answer.

9

[4 marks]

A prism has a cross-section that is a right-angled triangle with shorter sides 6 cm and 8 cm. The prism is 15 cm long.

(a) Work out the area of the cross-section.

(b) Work out the volume of the prism.

(c) Work out the total surface area of the prism.

10

[4 marks]

A shape is made from a rectangle 12 cm by 8 cm with a semicircle of diameter 8 cm cut out of one of the shorter ends.

Work out the area of the shape that is left, correct to 1 decimal place.

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Work out the sum of the interior angles of a 12-sided polygon.

2

[2 marks]

A cylinder has radius 4 cm and height 10 cm.

Work out its volume, correct to 3 significant figures.

3

[3 marks]

A right-angled triangle has hypotenuse 12 cm.

One of the other angles is 35° .

Work out the length of the side opposite the 35° angle, to 3 significant figures.

4

[3 marks]

In a right-angled triangle the side opposite angle θ is 7 cm and the side adjacent to θ is 9 cm.

Work out the size of angle θ , correct to 1 decimal place.

- 5 [4 marks]
- (a) A, B and C are points on a circle with centre O. The angle AOC at the centre is 130° , and B lies on the major arc. Work out the size of angle ABC, giving a reason.
- (b) PQRS is a cyclic quadrilateral. Angle PSR = 115° . Work out the size of angle PQR, giving a reason.
-

- 6 [4 marks]
- In triangle ABC, angle A = 42° , angle B = 68° and side a (opposite A) = 9 cm.
- Work out the length of side b, correct to 3 significant figures.
-

- 7 [4 marks]
- A triangle has two sides of length 7 cm and 9 cm with an angle of 58° between them.
- (a) Work out the length of the third side, to 3 significant figures.
- (b) Work out the area of the triangle, to 3 significant figures.
-

- 8 [4 marks]
- A sector of a circle has radius 9 cm and angle 140° .
- (a) Work out the length of the arc, to 3 significant figures.
- (b) Work out the area of the sector, to 3 significant figures.
-

- 9 [4 marks]
- In triangle OAB, the vector from O to A is **a** and the vector from O to B is **b**.
- M is the midpoint of AB. N is the point on OB such that ON : NB = 1 : 2
- (a) Write the vector from A to B in terms of **a** and **b**.
- (b) Write the vector from O to M in terms of **a** and **b**.
- (c) Write the vector from M to N in terms of **a** and **b**.
-

10

[5 marks]

A cuboid ABCDEFGH measures 8 cm by 6 cm by 5 cm, where ABCD is the 8 cm by 6 cm base and AE = 5 cm is vertical.

- (a) Work out the length of the diagonal AG that runs from one corner of the base to the opposite corner of the top. Give your answer to 3 significant figures.
- (b) Work out the angle that AG makes with the base ABCD, to 1 decimal place.